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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Machine Learning-Based Error Detection in Networks

COMPUTER NETWORKS (CSA0714)

TECHNICAL PRESENTATION

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INTRODUCTION

- ❑ Computer networks transfer data between devices.
- ❑ Network errors can cause packet loss and communication failures.
- ❑ Traditional methods detect errors after they occur.
- ❑ Machine Learning (ML) analyses network traffic and detects errors early.
- ❑ ML improves network reliability and performance.



Network Errors And Traditional Detection

- ❑ Common network errors: Packet Loss, Congestion, Link Failures.
- ❑ Traditional methods: Checksum, CRC, Parity Check.
- ❑ These methods detect errors after they occur.
- ❑ They cannot predict future network failures.
- ❑ **Key Takeaway :** *Traditional methods detect errors; Machine Learning predicts them.*

Machine Learning-Based Error Detection

- ❑ ML analyses network traffic data.
- ❑ Identifies abnormal patterns and anomalies.
- ❑ Detects errors before they affect network performance.
- ❑ Helps predict and prevent network failures.
- ❑ **Key Takeaway :** *Machine Learning enables intelligent and proactive error detection.*

Applications And OSI Integration

- ❑ Used in telecom networks, cloud computing, and data center .
- ❑ Detects errors across multiple OSI layers.
- ❑ Improves network reliability and efficiency.
- ❑ Reduces downtime and service interruptions.
- ❑ **Key Takeaway :** *ML enhances network performance across the 'OSI model.*

Advantages & Future Scope

- ❑ Faster and more accurate error detection.
- ❑ Reduces network downtime and maintenance costs.
- ❑ Improves reliability and user experience.
- ❑ Supports future technologies like AI-driven and self-healing networks.
- ❑ **Conclusion :** *Machine Learning makes networks smarter, more reliable, and ready for the future.*

THANK YOU